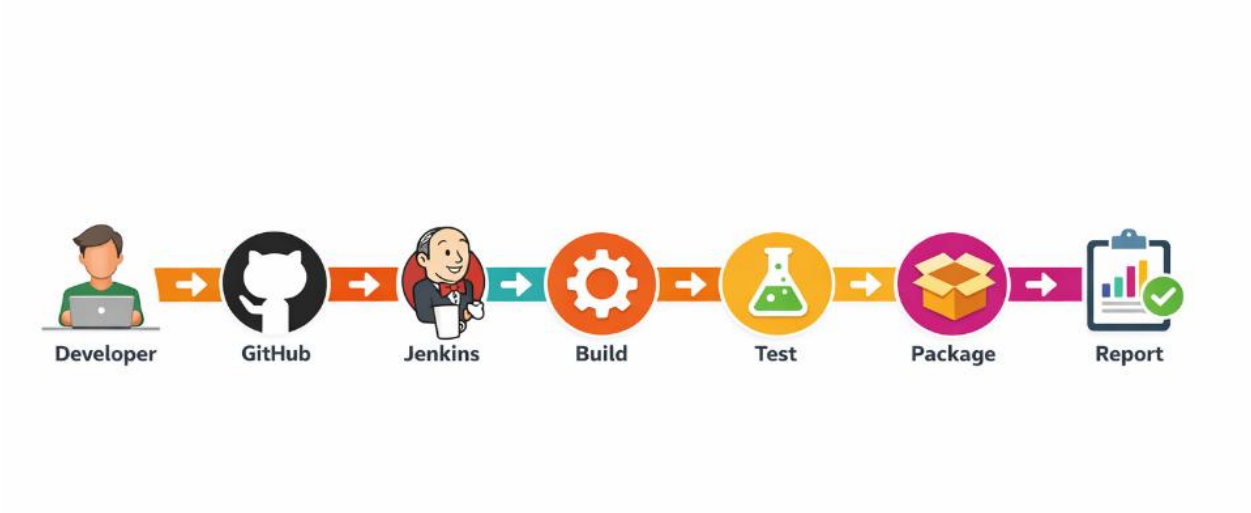
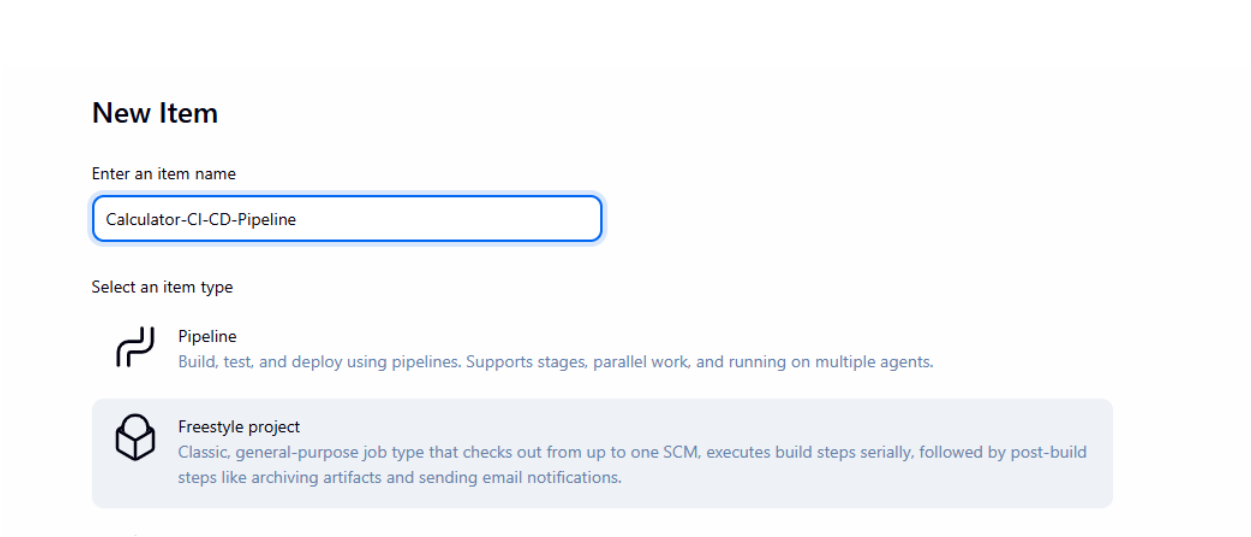
⭐ PROJECT: Jenkins CI/CD Pipeline for Java Calculator Application

Project Architecture:



First, we create a new Jenkins Freestyle Project:



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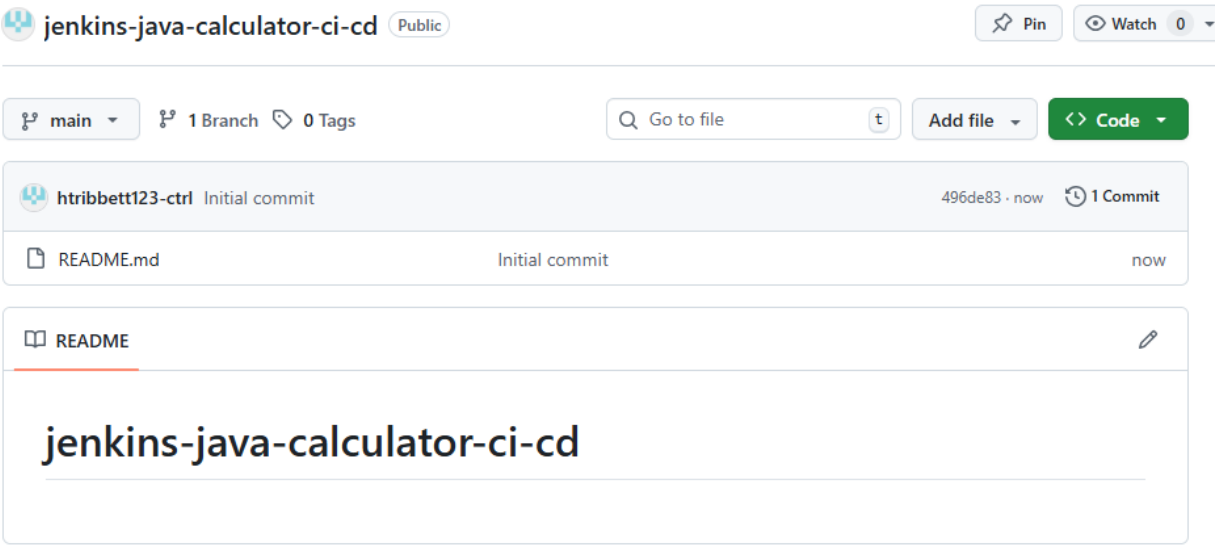
Create Your GitHub Repository



Am creating the folders necessary for this project if I paste in the file name:

src/main/java/com/example/calculator/Calculator.java – GitHub creates all the folder paths.

The code below is the actual application code of the calculator:

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jenkins-java-calculator-ci-cd / src / main / java / com / example / calculator / Calculator.java in main

Edit Preview

```java
package com.example.calculator;

public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        if (b == 0) {
            throw new IllegalArgumentException("Cannot divide by zero");
        }
        return a / b;
    }
}
```

src/test/java/com/example/calculator/CalculatorTest.java:

This code tests the application code. Jenkins runs this code through maven's surefire plugin:

```java
package com.example.calculator;


import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;
```

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```
public class CalculatorTest {

  Calculator calc = new Calculator();

  @Test
  void testAdd() {
    assertEquals(5, calc.add(2, 3));
  }

  @Test
  void testSubtract() {
    assertEquals(1, calc.subtract(3, 2));
  }

  @Test
  void testMultiply() {
    assertEquals(6, calc.multiply(2, 3));
  }

  @Test
  void testDivide() {
    assertEquals(2, calc.divide(4, 2));
  }

  @Test
```

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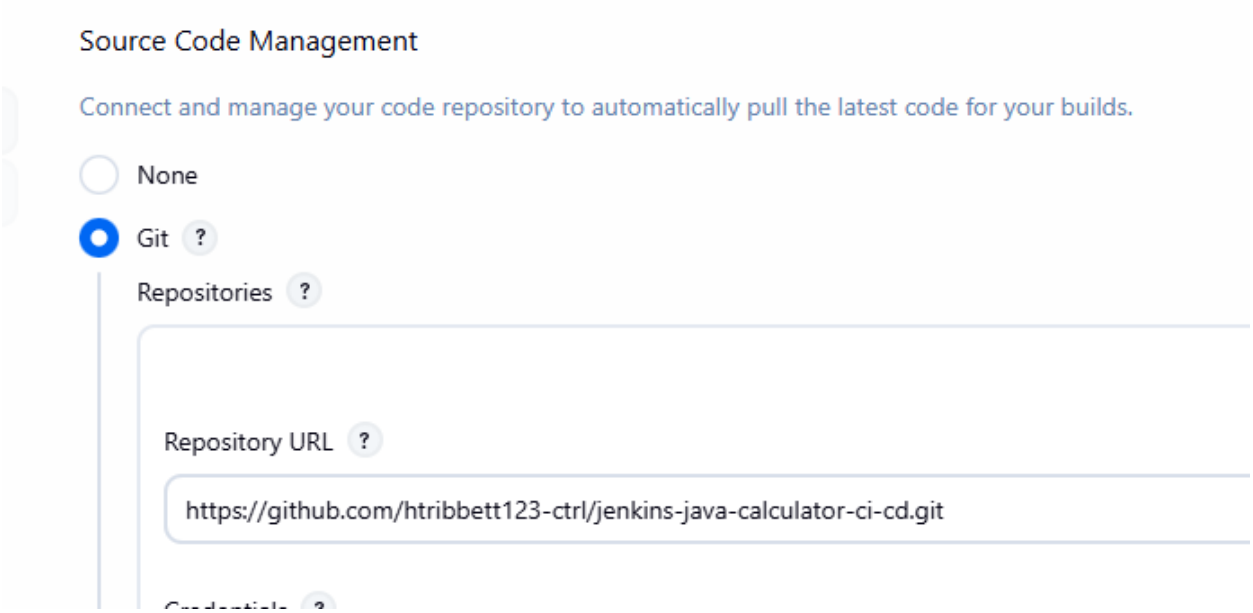
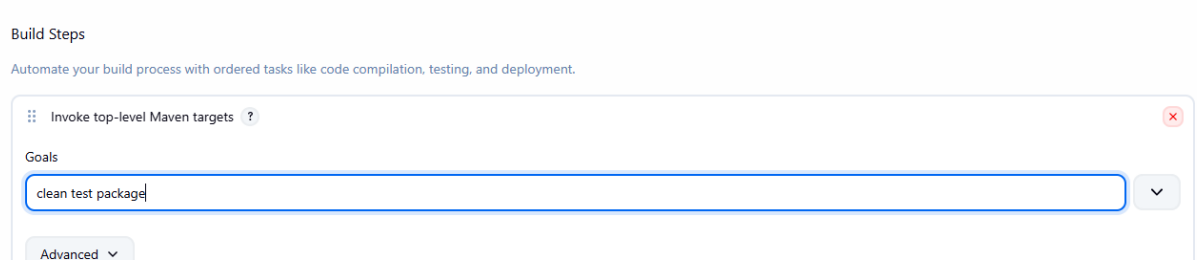
```
  void testDivideByZero() {

    assertThrows(IllegalArgumentException.class, () -> calc.divide(5, 0));

  }

}
```





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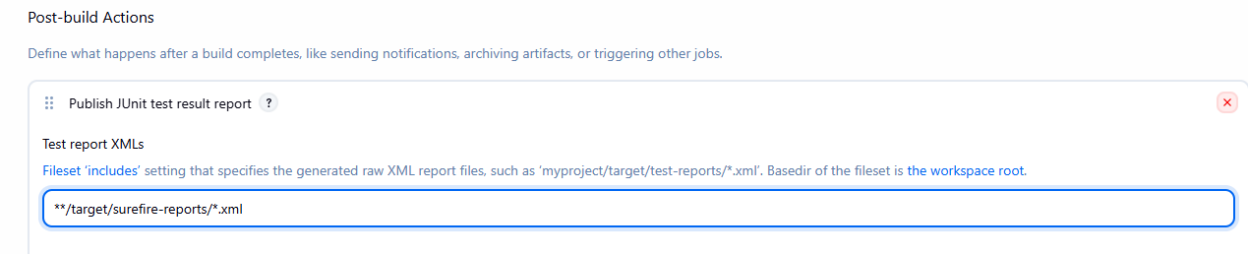
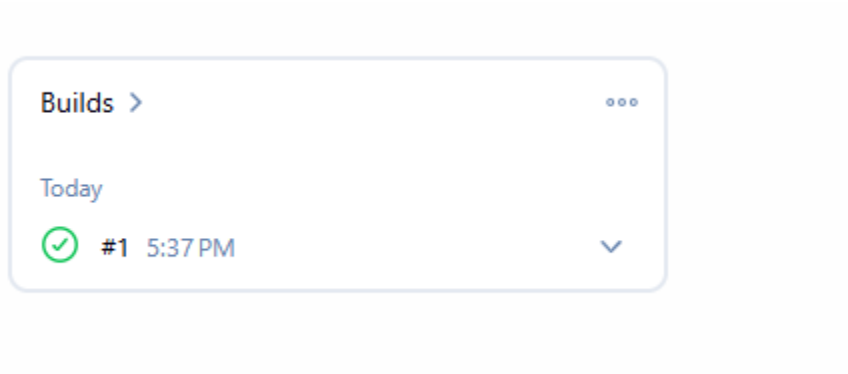
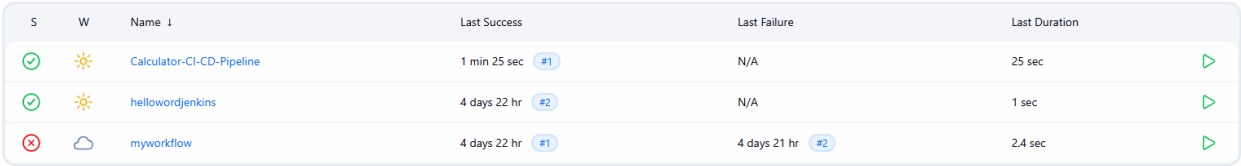
This makes test results appear in Jenkins:



Clicked save then build:





S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☼	Calculator-CI-CD-Pipeline	1 min 25 sec #1	N/A	25 sec
✓	☼	hellowordjenkins	4 days 22 hr #2	N/A	1 sec
✗	☁	myworkflow	4 days 22 hr #1	4 days 21 hr #2	2.4 sec

```
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-11/zstd-jni-1.5.5-11.jar (6.8 MB at 17 MB/s)
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\Calculator-CI-CD-Pipeline\target\calculator-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  19.587 s
[INFO] Finished at: 2026-03-08T17:38:12-07:00
[INFO] ------------------------------------------------------------------------
Recording test results
[Checks API] No suitable checks publisher found.
Finished: SUCCESS
```

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Workspace of Calculator-CI-CD-Pipeline on Built-In Node

Calculator-CI-CD-Pipeline /

- .git
- src
- target
- pom.xml Mar 8, 2026, 5:37:48 PM 1.76 KiB
- README.md Mar 8, 2026, 5:37:48 PM 1.42 KiB

Tests 5 ✓ 5 Took 0.11 sec

All tests are passing!

Nice one! All 5 tests are passing.

All Tests

Package	Failed	Skipped	Passed	Total	Duration
com.example.calculator	0	0	5 +5	5 +5	68 ms

Test ran: 5

CalculatorTest 5 ✓ 5 Took 68 ms

All Tests

Name	Age	Duration
testAdd	0	2 ms
testDivide	0	1 ms
testDivideByZero	0	62 ms
testMultiply	0	1 ms
testSubtract	0	2 ms